

Every word, unequal frequencies: digits of power Lambert series

For integers $b, r \geq 2$, consider

$$L_{b,r} = \sum_{m=1}^{\infty} \frac{1}{b^{mr} - 1}.$$

A real number is *disjunctive in base b* if every finite word over $\{0, \dots, b-1\}$ occurs in its fractional base- b expansion. It is *normal in base b* if every word of length ℓ has limiting frequency $b^{-\ell}$. Occurrences here may overlap, and frequencies are taken along all initial segments.

Theorem 1. *For every $b, r \geq 2$, every finite base- b word has a strictly positive limiting frequency in $L_{b,r}$, but $L_{b,r}$ is not normal in base b . In particular, it is disjunctive and irrational.*

More precisely, the empirical measures of $\{b^n L_{b,r}\}$ converge to an explicit probability measure $\nu_{b,r}$ on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. This measure is invariant and ergodic under $T_b(x) = bx \pmod{1}$, has no atoms, gives positive mass to every nonempty open set, and has Kolmogorov–Sinai entropy zero.

The proof has two parts. Congruences force a prescribed word while an average estimate controls the carry from later terms. Separately, a periodic approximation identifies the limiting distribution and shows that its entropy is zero. The arithmetic construction develops the Chowla–Erdős method as used by Duverney, Duverney–Tachiya, and Vandehey [1, 2, 3]. Irrationality of these constants already follows from [2, Corollary 1.2]; the issue here is their digit distribution.

1 Forcing a word

Fix $b, r \geq 2$ and write $L = L_{b,r}$. For $v \geq 1$, let

$$c_r(v) = \#\{d \geq 1 : d^r \mid v\} = \prod_{p^a \parallel v} \left(\left\lfloor \frac{a}{r} \right\rfloor + 1 \right).$$

Expanding each denominator as a geometric series gives

$$L = \sum_{v \geq 1} c_r(v) b^{-v}, \quad \{b^n L\} = \left\{ \sum_{j \geq 1} c_r(n+j) b^{-j} \right\}. \quad (1)$$

All rearrangements are justified by nonnegativity and $c_r(v) \leq v$. The coefficients need not be digits: this is why the carry matters.

Two elementary estimates

Lemma 1 (Means on a progression). *For positive integers u, Q ,*

$$\lim_{M \rightarrow \infty} \frac{1}{M} \sum_{m=0}^{M-1} c_r(u + Qm) = \sum_{d \geq 1} \frac{\gcd(d^r, Q)}{d^r} \mathbf{1}_{\gcd(d^r, Q) \mid u}.$$

In the Euler product on the right, primes not dividing Q contribute factors $(1 - p^{-r})^{-1}$. If $p^s \parallel Q$ with $s = r(B-1) + 1$ for an integer $B \geq 1$, its factor is at most $B+1$; it is 1 if $p \nmid u$.

Proof. The congruence $d^r \mid u + Qm$ is either insoluble or one residue class modulo $d^r / \gcd(d^r, Q)$. This gives the stated limit for each d . The average of its indicator is at most $(u + Q)/d^r$: the progression values are distinct positive integers bounded by $u + QM$. Since $\sum d^{-r} < \infty$, dominated convergence permits summation.

Multiplicativity gives the Euler product. If $p^s \nmid u$, the valuation at p is fixed below s , and its factor is at most B . If $p^s \mid u$, the factor is

$$B + \sum_{k \geq 1} p^{s-r(B-1+k)} = B + \frac{p}{p^r - 1} \leq B + 1.$$

If $p \nmid u$, the valuation is zero and the factor is 1. $\square$

The same domination allows passage through a weighted tail: for fixed u, Q, H , the mean of $\sum_{j > H} c_r(u + Qm + j)b^{-j}$ is the sum of its termwise means. Indeed, the summable majorant is $(u + Q + j)\zeta(r)b^{-j}$.

Lemma 2 (Power-free values). *If $u, v \geq 1$ and $\gcd(u, v) = 1$, then the indices $m \geq 0$ for which $u + vm$ is r -free have lower density at least $1/4$. Here r -free means divisible by no p^r with p prime.*

Proof. A prime dividing v divides no progression value. Every other prime excludes at most $M/p^r + 1$ of the first M indices. Only primes at most $\sqrt{u + vM}$ can exclude a value, since $r \geq 2$. Thus the excluded proportion is at most

$$\sum_p p^{-r} + O_{u,v}(M^{-1/2}) \leq \sum_{n=2}^{\infty} n^{-2} + O_{u,v}(M^{-1/2}) \leq \frac{3}{4} + O_{u,v}(M^{-1/2}).$$

The last inequality follows from $1/4 + \int_2^{\infty} x^{-2} dx = 3/4$. $\square$

The congruences and the carry

Let a prescribed word have length $\ell \geq 1$ and integer value $0 \leq w < b^\ell$. Its digit interval is $I_w = [w/b^\ell, (w + 1)/b^\ell)$. Put

$$t = \ell + 1, \quad A = bw + 1, \quad e = r(A - 1), \quad \eta = \frac{1}{2bt}.$$

The closed interval

$$K_w = \left[\frac{A}{b^t}, \frac{A}{b^t} + \eta \right] \quad \text{lies strictly inside } I_w. \quad (2)$$

The extra digit 1 leaves room for a small positive carry.

Choose a large integer $H \geq \max(t, 3)$; its size will be fixed below. For each $j \in \{0, \dots, H\} \setminus \{t\}$ choose H primes, all distinct and greater than $H^3 + H + 2$. Call these the primes assigned to j , let D be their total number, and put

$$s = r(b - 1) + 1, \quad Q = 2^{e+1} \prod_{p \text{ assigned}} p^s.$$

We will use only $D \leq H(H + 1)$. The Chinese remainder theorem gives $0 \leq a < Q$ satisfying

$$a + t \equiv 2^e \pmod{2^{e+1}}, \quad a + j \equiv p^{s-1} \pmod{p^s} \quad \text{for each prime } p \text{ assigned to } j.$$

Take $n = a + Qm$ with $m \geq 1$. Each prime assigned to j has exact valuation $r(b-1)$ in $n+j$, so contributes a factor b to $c_r(n+j)$. Consequently,

$$b^j \mid c_r(n+j) \quad (1 \leq j \leq H, j \neq t). \quad (3)$$

Write $n+t = 2^e u_m$. The u_m form an odd positive arithmetic progression of step $Q/2^e$. No assigned prime divides $n+t$: a prime assigned to j would then divide $t-j$, although $0 < |t-j| \leq H < p$. Thus the cofactor progression is coprime. By Lemma 2, u_m is r -free on a set of lower density at least $1/4$. At those indices,

$$c_r(n+t) = \left(\frac{e}{r} + 1\right) c_r(u_m) = A. \quad (4)$$

It remains to control $R_H(n) = \sum_{j>H} c_r(n+j)b^{-j}$. Let M_j be the mean of $c_r(a+Qm+j)$ on this progression, and put $C = 2(A+1)$. If $H < j \leq H^3$, no assigned prime divides $n+j$: subtracting its assigned shift gives a nonzero difference smaller than the prime. Lemma 1, together with $\zeta(r) \leq 2$, gives

$$M_j \leq C \quad (H < j \leq H^3), \quad M_j \leq C(b+1)^{H(H+1)} \quad (j > H^3).$$

The factor at 2 is at most $A+1$; each assigned prime costs at most $b+1$ when it is not excluded. Summing the two geometric tails yields

$$\lim_{M \rightarrow \infty} \frac{1}{M} \sum_{m=1}^M R_H(a+Qm) \leq \frac{C}{b-1} \left(b^{-H} + (b+1)^{H(H+1)} b^{-H^3}\right). \quad (5)$$

For $H \geq 3$, the second term in parentheses is at most b^{-H} : use $b+1 \leq b^2$ and $2H(H+1)+H \leq H^3$. Choose H so that the right side of (5) is less than $\eta/16$, and only then choose the primes. The estimate is independent of their sizes.

Markov's inequality shows that $R_H(n) \geq \eta$ on a set of upper density less than $1/16$ within the progression. Removing these indices from the power-free set leaves positive lower density. On the remaining indices, (1), (3), and (4) give

$$\{b^n L\} = \frac{A}{b^t} + R_H(n) \in K_w.$$

A set of positive lower density in one fixed progression has positive lower density among all integers. We have therefore proved:

$$\liminf_{N \rightarrow \infty} \frac{1}{N} \#\{0 \leq n < N : \{b^n L\} \in K_w\} > 0 \quad \text{for every word } w. \quad (6)$$

In particular L is disjunctive. It is irrational, since the radix orbit of a rational number is finite, whereas (6) makes this orbit dense in the circle.

2 The limiting distribution

The second part of the proof starts from a different expression for the same carry. For $d \geq 1$, put

$$g_d(n) = \frac{b^{n \bmod d}}{b^d - 1}, \quad W(n) = \sum_{m \geq 1} g_{m^r}(n), \quad W_U(n) = \sum_{m=1}^U g_{m^r}(n).$$

Summing the geometric series over j with $d \mid n+j$ shows that $W(n) = \sum_{j \geq 1} c_r(n+j)b^{-j}$. Thus $W(n) \pmod{1} = \{b^n L\}$.

An approximation valid for every prefix

The d values of g_d in a period increase from its minimum to its maximum and have total $1/(b-1)$. The average over an initial part of that period is therefore at most the complete-period average. Splitting any prefix into complete periods and one initial part gives

$$\frac{1}{N} \sum_{n=0}^{N-1} g_d(n) \leq \frac{1}{d(b-1)} \quad (N \geq 1).$$

Summing over $m > U$ proves the key estimate

$$\sup_{N \geq 1} \frac{1}{N} \sum_{n=0}^{N-1} |W(n) - W_U(n)| \leq \varepsilon_U := \frac{1}{b-1} \sum_{m>U} m^{-r} \rightarrow 0. \quad (7)$$

Each W_U is periodic, with period dividing $(U!)^r$. The uniformity in N lets us pass from these periodic sequences to the actual digit sequence.

A compact group model

Record all power residues at once:

$$G = \overline{\{(n \bmod m^r)_{m \geq 1} : n \in \mathbb{Z}\}} \subseteq \prod_{m \geq 1} \mathbb{Z}/m^r \mathbb{Z}.$$

This is a compact group. Let μ be its Haar probability measure and $R(x) = x + \mathbf{1}$, where $\mathbf{1} = (1 \bmod m^r)_{m \geq 1}$. Any finite collection of coordinates ranges over a single finite cyclic orbit. Haar measure projects to the uniform measure on that orbit. The rotation R is ergodic: an invariant L^2 function is invariant under the dense subgroup generated by $\mathbf{1}$, hence, by continuity of translations in L^2 , under every translation, and is constant almost everywhere. This is the usual compact-group rotation argument [4]. Related residue models in arithmetic dynamics appear in [5, 6].

For $x \in G$, let $\rho_m(x) \in \{0, \dots, m^r - 1\}$ represent its m th coordinate. Define

$$W(x) = \sum_{m \geq 1} \frac{b^{\rho_m(x)}}{b^{m^r} - 1}, \quad \Phi(x) = W(x) \pmod{1}, \quad \nu = \Phi_* \mu.$$

Every coordinate is uniform, so

$$\int_G W d\mu = \frac{\zeta(r)}{b-1} < \infty.$$

The series is therefore finite almost everywhere; define $\Phi = 0$ on its null exceptional set. Each summand satisfies

$$b \frac{b^{\rho_m(x)}}{b^{m^r} - 1} - \frac{b^{\rho_m(Rx)}}{b^{m^r} - 1} \in \mathbb{Z}.$$

Reduce finite partial sums modulo 1 and pass to the limit where both series converge. This proves

$$\Phi \circ R = T_b \circ \Phi \quad \mu\text{-almost everywhere.} \quad (8)$$

Thus ν is invariant and ergodic under T_b .

Let $W_U(x)$ be the corresponding finite sum on G . Its Haar tail satisfies $\int |W - W_U| d\mu = \varepsilon_U$. If f is a Lipschitz function on the circle, (7) bounds the difference between the integer averages of $f(W)$ and $f(W_U)$ by $\text{Lip}(f)\varepsilon_U$. The same bound holds for their Haar integrals. For fixed

U , the integer average of $f(W_U)$ converges to its Haar integral, because it runs periodically through the finite cyclic quotient. Hence

$$\limsup_{N \rightarrow \infty} \left| \frac{1}{N} \sum_{n < N} f(\{b^n L\}) - \int f d\nu \right| \leq 2 \text{Lip}(f) \varepsilon_U.$$

Letting $U \rightarrow \infty$, and using uniform approximation of continuous circle functions by Lipschitz functions, proves

$$\frac{1}{N} \sum_{n < N} \delta_{\{b^n L\}} \implies \nu. \quad (9)$$

This identifies the distribution of the specified orbit, not just of almost every point of the model.

From the distribution to word frequencies

For the closed interval K_w , weak convergence gives

$$\nu(K_w) \geq \limsup_{N \rightarrow \infty} \frac{1}{N} \# \{n < N : \{b^n L\} \in K_w\} > 0$$

by (6). Every nonempty open arc contains a digit interval, so ν has full support.

An ergodic invariant probability measure with an atom is supported on a finite periodic orbit. Indeed, atom masses cannot decrease under forward iteration, so the orbit of an atom must eventually repeat. The resulting cycle has positive measure and is invariant modulo a null set; ergodicity gives it full measure. Full support on the circle excludes this possibility. Thus ν has no atoms.

The boundary of I_w is finite and therefore has measure zero. Equation (9) now gives the limiting word frequency

$$\lim_{N \rightarrow \infty} \frac{1}{N} \# \{n < N : \{b^n L\} \in I_w\} = \nu(I_w) > 0.$$

3 Zero entropy and failure of normality

Here entropy measures the average information per symbol in a stationary sequence of finite labels. For a finite random variable X with probabilities p_i , its Shannon entropy is $H(X) = -\sum_i p_i \log p_i$, with $0 \log 0 = 0$. We prove directly that every finite measurable labeling of the group rotation has entropy rate zero.

Let $X : G \rightarrow \{1, \dots, k\}$ be such a labeling. Sets depending on finitely many coordinates generate the Borel sigma algebra of G . Consequently X can be approximated by a finite-coordinate labeling Y with mismatch probability $p = \mu(X \neq Y)$ as small as desired. For completeness, measurable sets approximable in measure by the finite-coordinate algebra form a sigma algebra; approximate the finitely many fibers of X and assign any overlaps in a fixed order. There is an integer $P \geq 1$ such that $Y \circ R^P = Y$.

Define the error label E to be $*$ when $X = Y$, and X otherwise. The first n labels of X along an orbit are determined by the first P labels of Y and the first n error labels. By the elementary reconstruction and subadditivity inequalities for Shannon entropy, and invariance of μ ,

$$H(X, X \circ R, \dots, X \circ R^{n-1}) \leq P \log k + nH(E).$$

Moreover,

$$H(E) \leq -p \log p - (1-p) \log(1-p) + p \log k \longrightarrow 0 \quad (p \rightarrow 0).$$

Divide by n , let $n \rightarrow \infty$, and then improve the approximation. Every finite labeling has entropy rate zero, so their supremum, the Kolmogorov–Sinai entropy of R , is zero. Iterating (8) outside a null set shows that pulling a circle labeling back through Φ preserves all its block distributions. Therefore T_b also has entropy zero with respect to ν .

If L were normal, its limiting word frequencies would give each of the b^n length- n digit intervals measure b^{-n} under ν . The digit labeling would then have block entropy $n \log b$, and hence positive entropy rate $\log b$. This contradicts the preceding result and completes the proof of Theorem 1.

Lean verification. The accompanying source package proves all conclusions of Theorem 1 in `PowerLambert.powerLambert_full_theorem`, with only the hypotheses $b, r \geq 2$. It uses Lean 4.34.0-rc2, Mathlib [7], and the Shannon entropy library from the Polynomial Freiman–Ruzsa formalization [8]; exact revisions and build instructions are recorded in the package. The axiom audit contains only Lean’s standard `propext`, `Classical.choice`, and `Quot.sound`. The formalization also proves nonnormality directly from periodic approximation; the paper uses the shorter entropy consequence above.

References

- [1] D. Duverney, *Arithmetical functions and irrationality of Lambert series*, AIP Conference Proceedings **1385** (2011), 5–16. [Author manuscript](#).
- [2] D. Duverney and Y. Tachiya, *Refinement of the Chowla–Erdős method and linear independence of certain Lambert series*, Forum Mathematicum **31** (2019), 1557–1566. [Author manuscript](#).
- [3] J. Vandehey, *On an incomplete argument of Erdős on the irrationality of Lambert series*, Integers **13** (2013), A58. [arXiv:1206.0340](#).
- [4] M. Mirzakhani, lectures; T. Feng, notes, *Introduction to Ergodic Theory* (2014). [Lecture notes](#).
- [5] F. Cellarosi and Ya. G. Sinai, *Ergodic properties of square-free numbers*, Journal of the European Mathematical Society **15** (2013), 1343–1374. [Journal article](#).
- [6] F. Cellarosi and I. Vinogradov, *Ergodic properties of k -free integers in number fields*, Journal of Modern Dynamics **7** (2013), 461–488. [arXiv:1304.0214](#).
- [7] The Mathlib contributors, *The Lean mathematical library*, revision `de2ef68216c6`. [Source repository](#).
- [8] The Polynomial Freiman–Ruzsa formalization contributors, *Polynomial Freiman–Ruzsa conjecture formalization*, Shannon entropy library, revision `3d7898164ebf`. [Source repository](#).

Text licensed under [CC BY 4.0](#). The accompanying Lean code is licensed under Apache 2.0.